

Computer Graphics

Chapter 1 : Introduction

0xSaad Chapters

Computer Graphics

Computer Graphics : Creation , manipulation
and storing an images by computer

Computer Graphics

Image : combination of objects , light , viewer
(or camera)

Represents visual informations
and stored in photographic
films , canvas and paintings or
maybe digitally



Computer Graphics

Image Synthesis: or Image generation is construction of image from scratch

Rendering : Converts a 3D Model to 2D Image

Modeling : Models a 2D Image as 3D Model

Computer Graphics

CG Applications :

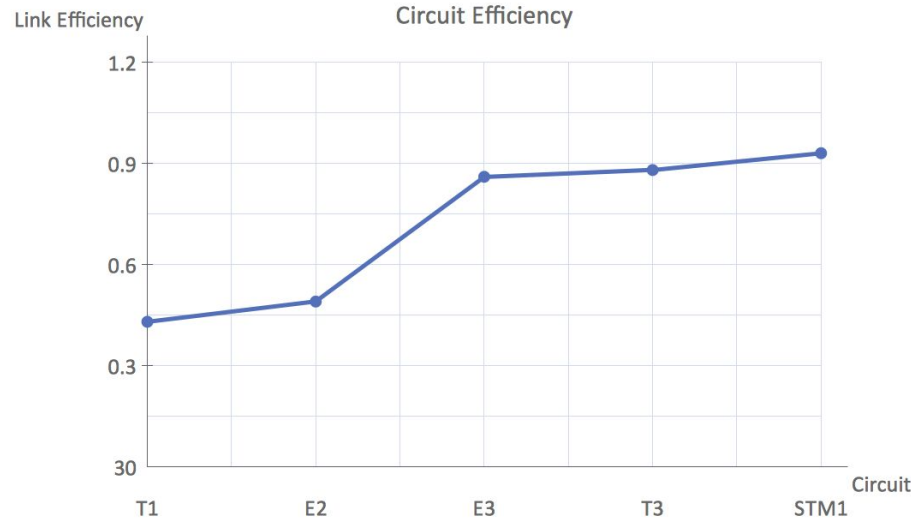
- Data Visualization



Computer Graphics

CG Applications :

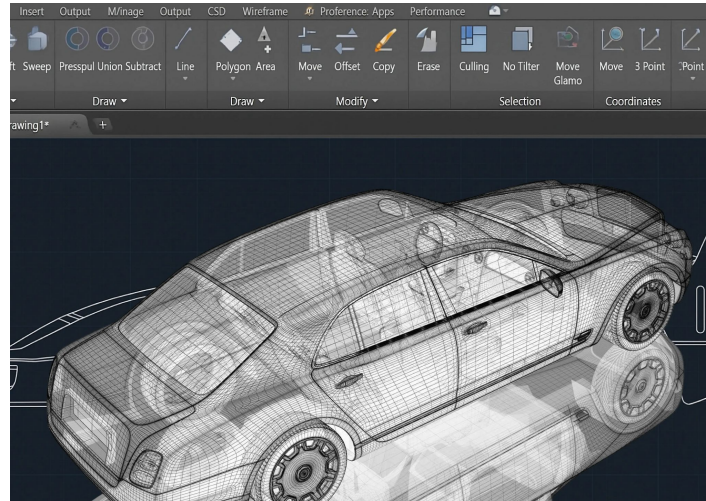
- Graphs and charts



Computer Graphics

CG Applications :

- Computer Aided Drawing (CAD)



Computer Graphics

CG Applications :

- Video Games



HL2 LOL

Computer Graphics

Computer Vision vs Computer Graphics :

- Computer Graphics : Creating and storing 2D and 3D Models and images using algorithms
- Computer Vision : Extracts useful informations from images and analyzing them.

Computer Graphics

Interactive vs Non Interactive :

- Interactive CG allows real-time user interaction with graphics environment (Video games - VR)
- Non Interactive CG does not respond to user (Ads - Movies - Static images)

Computer Graphics

CPU Vs GPU :

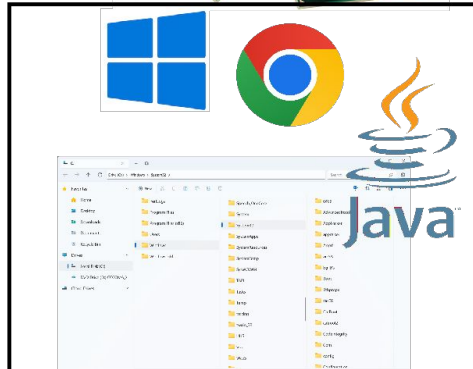
CPU : Brain of the computer , General-purpose processor used Sequential processing

GPU : Graphics specialized processor used Parallel processing.

Computer Graphics

CPU Vs GPU :

CPU works for general things
and all of time



GPU works for rendering
images in your screen



Computer Graphics

Pixels : Picture elements

Primitives : Basic elements of images
(Lines-Points and Polygons)

Resolution : Number of pixels per unit video display.

Computer Graphics

GPU Converts primitives to pixels

Primitives can be :

- Point



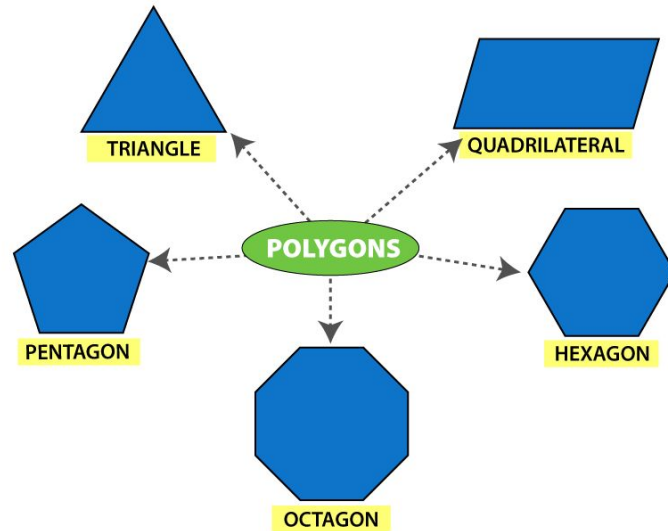
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- Line : connection between 2 points from source to destination



Computer Graphics

- Polygon : 2 Dimensional geometric shape can be circle , square , triangle or any forms



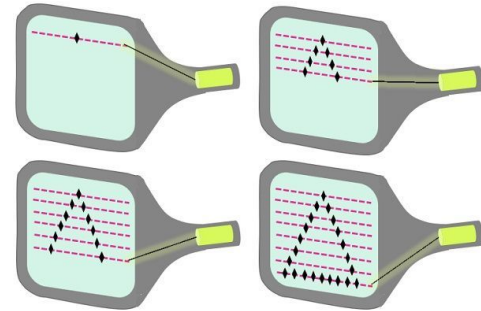
Computer Graphics

Raster Display Scan , Raster : rectangular array of dots or points

Uses primitives based on mathematical equations.

used in various types of display technologies :

- CRT
- LCD
- LED



Raster Scan: A raster scan system displays an item as a group of separate points along each screen line

Computer Graphics

Cathode Ray Tube : Widely used in old computers and TVs

uses:

- Vacuum tube
- Electron beam
- Phosphor coating

Computer Graphics

Color CRT Methods :

Beam Penetration : Red + Green phosphors

And only four colors :

- Red
- Blue
- Yellow
- Orange

Computer Graphics

Refresh Rate : complete images (frames) drawn on the screen in 1 second.

Frame Rate : Time between consecutive frames.

Resolution: a 2D term , maximum number of points displayable without overlap.

Computer Graphics

- **LCD** : Using liquid crystals , Requires backlight , Thin and light.
- **Plasma** : Large viewing angle , Bright display but Expensive and Need high power
- **LED** : is LCD but with LED Backlight
- **OLED** : No Backlight required

Computer Graphics

LED and OLED :



Computer Graphics

Basic algorithms needed :

- Transformation
- Hidden Surface Removal
- Rasterization

Computer Graphics

Two Dimensions : Flat Images (x,y)



Three Dimensions : Added depth (x,y,z)



Computer Graphics

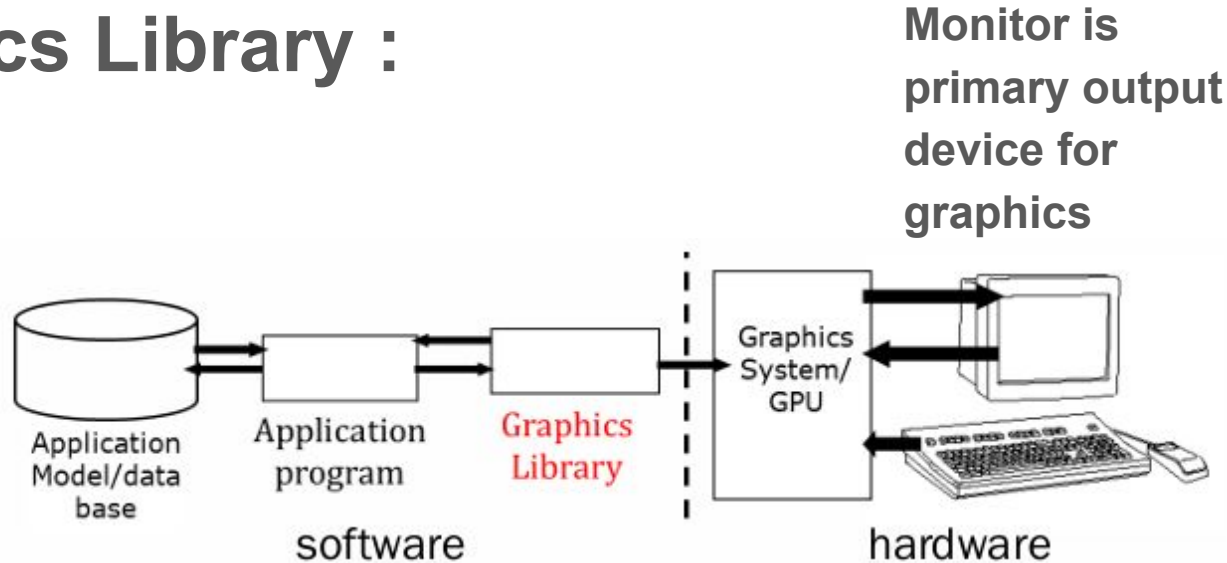
Graphics Library : Intermediary between Application and GPU , Draw primitives , Apply transformations and Handle lighting

Examples : OpenGL - DirectX - Vulkan



Computer Graphics

Graphics Library :

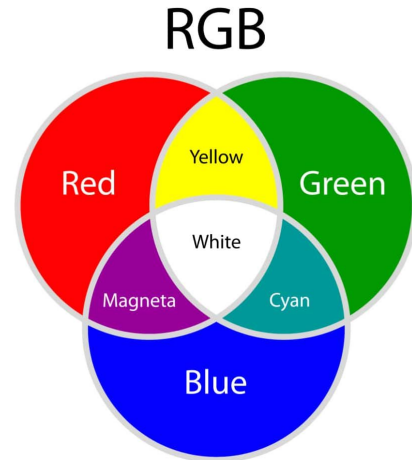


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Color Models :

RGB (Red Green Blue) : Model using red , green and blue patterns for make colors

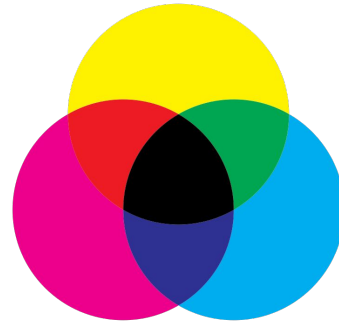
**Mainly used in
modern
applications**



Computer Graphics

Color Models :

CMYK : a subtractive color model , Combination of cyan , magenta , yellow and black



Computer Graphics

CMYK :

